

Notes: Brown, Fazzari, and Petersen (JF, 2009)
**Financing Innovation and Growth: Cash Flow, External Equity, and the 1990s
R&D Boom**

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1 Big picture

- **Question.** Can shifts in the supply of internal and external equity finance explain a large share of the dramatic U.S. R&D boom of the 1990s and the subsequent decline after 2000?
- **Setting.** The paper studies publicly traded U.S. firms in seven high-tech industries from 1990 to 2004, using Compustat panel data and a dynamic R&D-investment model.
- **Why this question matters.** The finance-and-growth literature shows that finance and growth move together, but it is harder to identify the concrete channel. This paper focuses on R&D, the key innovative input in endogenous-growth models.
- **Core empirical challenge.** A positive correlation between finance variables and R&D could reflect financing constraints, but it could also reflect investment demand, technological opportunities, or stock-market mispricing that makes firms want to appear more innovative.
- **Main conceptual contribution.** The paper argues that the relevant marginal financing source for young high-tech firms is not debt but *equity*: first internal cash flow, then public stock issuance. That is what makes external equity an important omitted channel in earlier work.
- **Main empirical strategy.** Estimate a dynamic Euler-equation-style R&D model and compare firms that are *young* (public for 15 years or less) with *mature* firms (public longer than 15 years). If financing constraints matter, finance variables should predict R&D only for the young firms that rely on equity at the margin.
- **Main baseline result.** For young firms, both contemporaneous cash flow and contemporaneous new stock issues have large, positive, and statistically significant effects on R&D. For mature firms, the same variables have coefficients near zero and are generally insignificant.
- **Main quantitative result.** Using the estimated young-firm coefficients, finance booms explain about 72% of the 1994–2000 R&D boom in the seven high-tech industries, and the finance bust explains about 78% of the 2001–2004 decline.
- **Key descriptive result.** The aggregate R&D boom is almost entirely a young-high-tech phenomenon. Mature firms do not exhibit a comparable boom in R&D, even though they also experience rising valuations and some increase in cash flow.
- **Main credibility result.** The young-firm finance effects survive industry-year dummies, Tobin's q controls, forecasting tests for future profitability, alternative age cutoffs, alternative sample-split criteria, alternative instrument sets, orthogonal deviations, and system GMM.
- **Economic takeaway.** Equity finance matters for innovation. Stock-market booms and cash-flow booms can relax financing constraints for young R&D-intensive firms and materially affect aggregate innovative activity.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- The late-1990s U.S. R&D boom is unusually large: privately financed industrial R&D rises from 1.40% of GDP in 1994 to 1.89% in 2000, then falls back to roughly 1.70% on average in

2002–2004.

- The boom is highly concentrated. Seven three-digit SIC industries account for virtually all of the 1994–2004 R&D cycle: drugs (283), office and computing equipment (357), communications equipment (366), electronic components (367), scientific instruments (382), medical instruments (384), and software (737).
- Young public firms in these industries finance R&D almost entirely with internal cash flow and public equity, not debt. That makes the financing channel especially visible in the data.
- Public firms undertake nearly all U.S. industrial R&D in this period: the Compustat public-firm total equals about 90.1% of National Science Foundation industrial R&D in 2003.

Interpretation. This is not a general story about all firms. It is a story about a sector where R&D is central, debt is a poor financing instrument, and public equity issuance is quantitatively important.

2.2 Data source and sample construction

- Main source: Compustat panel data for publicly traded firms, 1990–2004.
- Keep firms in the seven high-tech SIC industries listed above.
- Exclude firms incorporated outside the United States.
- Exclude firms with no stock price data.
- Exclude firms with fewer than six R&D observations.
- Exclude firms whose *sum* of cash-flow-to-assets over the sample is nonpositive.
- Trim all key variables at the 1% tails.

After these restrictions, the regression sample contains 1,347 firms and captures more than 90% of public-firm R&D in the high-tech industries.

Interpretation. The sample is designed to focus on the economically important public high-tech sector while avoiding very small startup firms whose ratios are extremely noisy and whose aggregate R&D weight is tiny.

2.3 Young versus mature firms

A firm’s “age” is measured as years since its first stock price appears in Compustat, which is typically the IPO year.

- **Young firms:** public for 15 years or less.
- **Mature firms:** public for more than 15 years.

The paper also checks alternative cutoffs of 10 and 20 years, with very similar conclusions.

Young firms are quantitatively important in the aggregate R&D cycle:

- the young-firm share of regression-sample R&D averages 33.8% over the sample period;

- it rises from 21.7% at the low end to 45.7% in 1998;
- it then falls to 26.1% by 2004.

Interpretation. The age split is not cosmetic. Young firms account for most of the cyclical action in R&D, and the financing story is built around that fact.

2.4 Key variables

All variables are scaled by beginning-of-period total assets.

- rd_t : research and development expense divided by beginning assets.
- $capex_t$: capital expenditures divided by beginning assets.
- s_t : net sales divided by beginning assets.
- cf_t : *gross cash flow*, defined as after-tax income before extraordinary items plus depreciation and amortization plus R&D expense, divided by beginning assets.
- stk_t : net cash raised from stock issues, equal to sales of common and preferred stock minus purchases of common and preferred stock, divided by beginning assets.
- dbt_t : net new long-term debt, equal to long-term debt issuance minus long-term debt reduction, divided by beginning assets.

The paper also reports financing shares relative to total finance raised, where total finance means cash flow plus new stock issues plus new debt.

2.5 What the financing mix looks like

Table I is one of the most important objects in the paper because it shows why young and mature firms should respond differently to finance shocks.

- R&D is large relative to physical investment for everyone, but especially for young firms.
- Mean rd_t is 0.194 for young firms and 0.098 for mature firms.
- Mean $capex_t$ is only 0.069 for young firms and 0.049 for mature firms.
- Mean sales-to-assets ratios are fairly similar across groups (1.227 for young versus 1.167 for mature firms).
- Mean cash flow is slightly higher for young firms (0.217 versus 0.172), but for young firms cash flow is still much smaller than the sum of R&D and capital spending.
- Mean stock issues are huge for young firms (0.268) and tiny for mature firms (0.021).
- New debt finance is near zero for both groups.

Financing shares sharpen the point:

- for young firms, average finance comes 65.9% from cash flow, 32.0% from public equity, and only 1.5% from debt;

- for mature firms, average finance comes 91.6% from cash flow, 6.8% from public equity, and 1.9% from debt.

Interpretation. Mature firms usually fund R&D internally. Young firms do not. They often exhaust internal cash flow and must rely on public equity as their marginal financing source.

2.6 Descriptive facts about the boom and bust

The paper documents the boom–bust pattern in both aggregate and sample data.

- In the full U.S. public-firm data, the seven high-tech industries account for virtually all of the economywide R&D cycle between 1994 and 2004.
- For young high-tech firms, the authors fit a geometric trend to real R&D over 1980–1993 and find a trend annual growth rate of 11.8%.
- For mature firms, the corresponding trend growth rate is 7.5%.
- From 1996 through 2001, actual young-firm R&D is on average 65% above its already-steep pre-1994 trend.
- By 2003 and 2004, young-firm R&D falls below trend.
- Mature-firm R&D after 1993 grows almost exactly along trend.

The financing variables show the same boom–bust pattern for young firms:

- mean young-firm stock issues-to-assets are 0.351 in 1995–2000 and only 0.070 in 2001–2004;
- mean young-firm cash flow-to-assets is 0.246 in 1995–2000 and 0.136 in 2001–2004.

Interpretation. The visual and descriptive evidence already suggests a finance-driven story: young-firm R&D, cash flow, and stock issuance rise together in the boom and fall together in the bust.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper’s financing hierarchy for R&D is the core economic idea.

- Debt is a poor financing instrument for young R&D-intensive firms because R&D has little collateral value, returns are highly uncertain, and financial distress is especially costly when value depends on future growth options.
- Internal equity (cash flow) is the lowest-cost source of finance.
- Once internal funds are exhausted, firms issue public equity.

- The supply curve for external equity is upward sloping because of flotation costs and asymmetric-information premia.

This leads to two key predictions:

- For **young firms**, a rightward shift in the supply of either internal cash flow or external equity should lower the marginal cost of funds and increase R&D.
- For **mature firms**, which often have internal funds in excess of desired investment and do not rely on equity issuance, the same supply shifts should have little effect on R&D.

The paper also expects finance coefficients to be well below one. Even constrained firms will not turn an extra dollar of finance into a full dollar of R&D, because firms have other uses of funds and R&D itself has high adjustment costs.

3.2 Baseline Euler-equation specification

The starting point is a dynamic Euler-equation model adapted from Bond and Meghir. In the absence of financing constraints, the empirical specification is

$$rd_{j,t} = \beta_1 rd_{j,t-1} + \beta_2 rd_{j,t-1}^2 + \beta_3 s_{j,t-1} + \beta_4 cf_{j,t-1} + d_t + \alpha_j + \nu_{j,t}.$$

Here:

- $rd_{j,t}$ is the R&D-to-assets ratio,
- $s_{j,t}$ is sales-to-assets,
- $cf_{j,t}$ is gross cash flow-to-assets,
- d_t is a time effect,
- α_j is a firm fixed effect.

Why scale by total assets instead of the R&D stock? Because the true R&D stock is hard to measure: perpetual-inventory methods require long histories and a depreciation rate for an intangible stock that is hard to pin down.

Under the structural model without financing constraints:

- β_1 should be positive and slightly above one;
- β_2 should be slightly less than negative one;
- β_3 should be positive under imperfect competition;
- β_4 should be negative because cash flow proxies for the cost of other factors of production in the Euler equation.

Interpretation. This baseline specification is meant to capture optimal dynamic R&D behavior in frictionless capital markets while controlling for expectations and firm dynamics.

3.3 Financing-augmented specification

To allow for financing constraints, the paper adds current and lagged finance variables:

$$\begin{aligned}rd_{j,t} = & \beta_1 rd_{j,t-1} + \beta_2 rd_{j,t-1}^2 + \beta_3 s_{j,t} + \beta_4 s_{j,t-1} \\ & + \beta_5 cf_{j,t} + \beta_6 cf_{j,t-1} + \beta_7 stk_{j,t} + \beta_8 stk_{j,t-1} + d_t + \alpha_j + \nu_{j,t}.\end{aligned}$$

Why add these terms?

- Current cash flow measures the contemporaneous supply of internal equity.
- Current stock issuance measures the contemporaneous supply of external equity.
- Current sales is added as an extra demand control, so the current cash-flow coefficient is not simply picking up contemporaneous product demand.

Prediction:

- for unconstrained or mature firms, the added finance variables should not matter much;
- for constrained young firms, cf_t and stk_t should enter positively.

Interpretation. Once these finance variables enter positively for young firms but not mature firms, the regression moves from a dynamic investment equation to a direct test of financing constraints on R&D.

3.4 Panel structure, GMM, and instruments

The preferred estimator is one-step Arellano–Bond GMM in first differences.

- First differencing removes firm fixed effects.
- All right-hand-side variables are treated as potentially endogenous.
- Baseline instruments are lagged levels dated $t - 3$ and $t - 4$, which remain valid if the original error has an MA(1) structure.
- Robustness checks add $t - 2$ instruments when the error is i.i.d.
- One-step GMM is preferred because two-step GMM standard errors are downward biased in small samples.

All standard errors are robust to heteroskedasticity and arbitrary within-firm serial correlation.

Time effects are handled in two ways:

- aggregate year dummies;
- three-digit-industry year dummies.

Interpretation. The industry-year dummies are important because they absorb time-varying technological opportunities and industry demand shocks. If finance variables still matter after those controls, the supply interpretation becomes much stronger.

3.5 Main regression results

Pooled sample. When all firms are pooled together, both cash flow and stock issues enter positively and significantly in the financing-augmented specification. But the pooled result hides strong heterogeneity.

Mature firms.

- Current cash flow: -0.015 with aggregate time dummies and -0.005 with industry-year dummies.
- Current stock issuance: 0.031 and 0.034 .
- The sums of current and lagged cash-flow coefficients are not significantly different from zero.
- The sums of current and lagged stock-issue coefficients are also not significantly different from zero.

Young firms.

- Current cash flow: 0.166 with aggregate time dummies and 0.150 with industry-year dummies.
- Current stock issuance: 0.148 in both versions.
- Both cash-flow and stock-issue effects are highly significant.
- Lagged finance terms are small, implying that the economically important effects are contemporaneous.

Interpretation. The finance effect is not just statistically present for young firms; it is economically meaningful and very similar across internal and external equity. That is exactly what one would expect if funds are fungible inside the firm once raised.

3.6 Economic interpretation of the coefficients

The paper summarizes the economic content of the young-firm estimates this way:

- a young firm invests roughly 15 cents of an additional dollar of equity finance in R&D;
- cash flow and stock issues have similar coefficients, consistent with internal and external equity being alternative sources of funds;
- the coefficient is much smaller than one because firms must also fund working capital and physical investment, and because R&D has high adjustment costs and is smoothed over time.

Interpretation. The coefficients are not meant to say “an extra dollar of cash flow creates 15 cents of social R&D forever.” They are marginal short-run partial-equilibrium responses inside a dynamic adjustment model.

3.7 Quantitative implications for the boom and bust

The paper translates the young-firm coefficients into boom arithmetic.

- Define “boom” R&D for 1994–2000 as actual young-firm R&D minus the projection of the 1980–1993 trend.
- Define analogous boom measures for cash flow and new stock issues.
- Using the young-firm coefficients from the industry-dummy specification, the combined cash-flow and stock-issue booms explain 72% of the 1994–2000 R&D boom.
- The finance bust explains 78% of the 2001–2004 fall in R&D.
- Cash flow accounts for just under two-thirds of the explained variation, with new stock issues accounting for the remainder.

Interpretation. This is the paper’s bridge from micro coefficients to the macro innovation cycle. The claim is not just that finance matters statistically, but that it matters quantitatively for aggregate R&D.

3.8 Was late-1990s R&D excessive?

A natural alternative story is that young firms overinvested in R&D to cater to irrationally exuberant equity markets. The paper argues against that interpretation.

- Mature firms also enjoyed rising valuations: average Tobin’s q rises by 78% from 1990 to 1999 for mature firms, versus 122% for young firms. Yet mature firms show no strong finance effect on R&D.
- If firms were simply trying to impress investors, physical investment should rise at least as much as R&D. Instead, the median capital-investment-to-assets ratio rises only 23.6% from 1990 to its peak, whereas the R&D-to-assets ratio rises 42.6%.
- In capital-spending regressions, the stock-issue coefficient for young firms is only 0.045, about 30% of the corresponding R&D coefficient.
- The finance–R&D relation is not stronger in weak-governance firms and not stronger for firms with less-experienced managers.

Interpretation. The paper does not deny that stock-market mispricing may have shifted the supply of external equity. What it denies is that the observed finance–R&D relation is best read as wasteful catering or overinvestment.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimate

The identification comes from three layers combined together:

- within-firm changes over time, because firm fixed effects are removed;
- common time-shock controls, especially industry-year dummies;

- cross-group heterogeneity between young and mature firms.

The key logical step is comparative: if finance variables were only proxying for general demand or opportunity shocks, they should matter for mature firms too. They do not.

4.2 Why reverse causality is relatively weak

Several features of the setting reduce the concern that current R&D is simply causing the finance variables:

- R&D has a long gestation period—the paper cites 4 to 6 years before it becomes productive—so short-run cyclical demand is a weak explanation for an aggregate R&D boom.
- The huge movements in cash flow and external equity issuance line up with broad profit and Nasdaq cycles that plausibly shift finance supply.
- Debt is essentially absent, so the usual issue that debt finance is an omitted substitute is muted here.

Interpretation. The paper is not claiming perfect exogeneity of cash flow or equity issuance. It is claiming that, in this particular sector and after controls, the remaining young-firm-only finance effect is hard to reconcile with a pure demand story.

4.3 Expectations and technological-opportunity controls

The paper takes expectations seriously.

- The Euler-equation approach is explicitly chosen because it nets out unobservable expectations under the maintained dynamic-optimization structure.
- The regressions include aggregate or industry-specific year dummies.
- Adding beginning-of-period Tobin's q leaves the finance results essentially unchanged.
- Even adding end-of-period Tobin's q —an especially demanding control because it contains all contemporaneous market information—does not materially change the finance coefficients.

A further test is very instructive:

- reduced-form forecasting regressions show that cash flow predicts future profitability *better for mature firms than for young firms*;
- yet mature-firm R&D shows no strong cash-flow sensitivity.

Interpretation. This is an important credibility point. If cash flow were merely a proxy for expected future profits, one should have expected the strongest effect where that predictive power is strongest—namely, among mature firms. The opposite occurs.

4.4 Why the young–mature comparison is central

The paper's interpretation depends heavily on the fact that the financing hierarchy differs sharply across the two groups.

- Young firms must use public equity on the margin because internal funds are not enough.
- Mature firms usually fund R&D internally and make little use of new equity.
- The aggregate R&D boom itself is confined almost entirely to young firms.

Interpretation. The age split gives the paper something close to an internal control group: mature firms face the same macro shocks, but not the same binding financing constraints.

4.5 Alternative sample splits tell the same story

The paper also splits firms using more traditional “constrained versus unconstrained” proxies.

- Firms with no bond rating or low payout ratios behave like the young-firm group: finance coefficients are positive and significant.
- Bond-rated or high-payout firms behave like the mature-firm group: finance coefficients are small and insignificant.

Interpretation. The results are not an artifact of one arbitrary age cutoff. The broader pattern is always the same: the firms most likely to face financing constraints are the ones whose R&D responds to equity finance.

4.6 Robustness to estimation method and instrument strength

The paper carefully addresses the worry that lagged levels may be weak instruments.

- Using $t - 2$ through $t - 4$ levels as instruments yields results very similar to the baseline.
- Orthogonal-deviations GMM also gives nearly identical finance coefficients.
- System GMM produces similar point estimates, although the Sargan test rejects the additional moment restrictions for the young-firm regressions, so the paper does not rely on that estimator.

Interpretation. The message is robustness, not estimator shopping. The finance result for young firms is stable across reasonable dynamic-panel implementations.

4.7 Negative-cash-flow firms and firms outside high tech

Two extra checks reinforce the interpretation.

Negative-cash-flow firms.

- These firms are tiny startups with extremely volatile ratios.
- They account for only 0.8%, 3.2%, and 4.3% of aggregate high-tech R&D in 1990, 1997, and 2004.
- Their cash-flow coefficient is negative, while their stock-issue coefficient is positive and significant (0.079), exactly as one would expect for firms that have not reached the stage where internal funds matter.

Young firms outside high tech.

- Their median R&D-to-assets ratio is only 0.030.
- Median cash flow exceeds the sum of median R&D and capital-spending ratios.
- Finance coefficients for these firms are small and mostly insignificant.

Interpretation. The finance–R&D channel is not a generic feature of all young public firms. It is strongest exactly where theory says it should be: young, R&D-intensive, equity-dependent high-tech firms.

4.8 What the paper can and cannot distinguish

What it can establish well.

- Finance matters for young high-tech firms' R&D.
- Internal and external equity both matter.
- These effects are quantitatively large enough to matter for aggregate R&D.

What it cannot fully pin down.

- It does not separately identify the exact primitive source of the external-equity supply shift (for example, pure mispricing versus other shifts in capital-market conditions).
- It does not provide a structural welfare decomposition of the socially optimal level of R&D.
- It does not prove that all of the late-1990s R&D rise was efficient, only that the evidence fits relaxed financing constraints much better than the main alternatives.

4.9 Relation to the literature

The paper sits at the intersection of three literatures:

- the financing-constraints literature, especially investment–cash-flow sensitivities;
- the R&D finance literature, which had largely focused on internal funds and had little to say about aggregate R&D cycles;
- the macro finance-and-growth literature, which emphasizes financial development but had less direct micro evidence on the innovation channel.

Its distinctive contribution is to show, with micro data, that equity finance supply shifts can move aggregate innovative activity in a quantitatively important way.

5 Tables and figures

5.1 Figure 1: the aggregate R&D boom is basically a high-tech boom

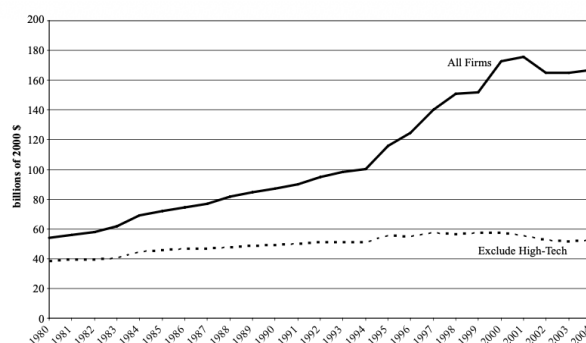


Figure 1. R&D investment. The solid line plots the sum of R&D for all publicly traded companies with coverage in Compustat (financial firms and utilities are excluded) over time. The dashed line plots the sum of R&D for firms in all industries *except* the seven high-tech industries with SIC codes 283, 357, 366, 367, 382, 384, and 737.

Read:

- Total public-firm R&D rises sharply from the mid-1990s to 2000.
- Once the seven high-tech industries are excluded, the boom almost disappears.
- The high-tech share of economywide public-firm R&D becomes dominant by the late 1990s.

Economic message. The paper is not trying to explain every form of investment. It is explaining a very specific and very important innovation boom concentrated in high-tech sectors.

5.2 Figures 2a–2c: young firms' R&D, cash flow, and stock issuance move together

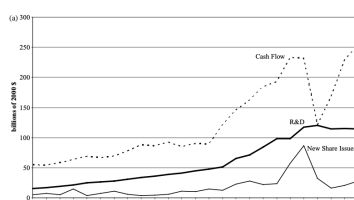


Figure 2a. High-tech R&D, cash flow, and new share issues (all firms). The sample is all publicly traded companies in high-tech industries 283, 357, 366, 367, 382, 384, and 737 with coverage in Compustat. The heavy line plots the sum of R&D for all high-tech firms, the dashed line plots the sum of gross cash flow, and the thin line plots the sum of net new stock issues with negative net issues set equal to zero.

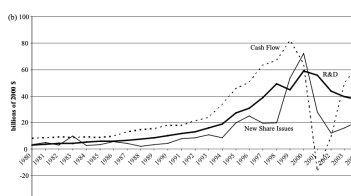


Figure 2b. High-tech R&D, cash flow, and new share issues (young firms). The sample is all young high-tech firms with coverage in Compustat. A firm is classified as young for the first 15 years following the year it first appears in Compustat with a stock price. The high-tech industries are SICs 283, 357, 366, 367, 382, 384, and 737. The heavy line plots the sum of R&D for all young high-tech firms, the dashed line plots the sum of gross cash flow, and the thin line plots the sum of net new stock issues with negative net issues set equal to zero.

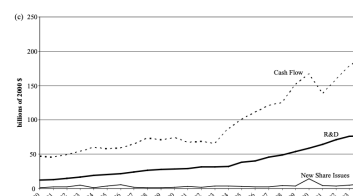


Figure 2c. High-tech R&D, cash flow, and new share issues (mature firms). The sample is all mature high-tech firms with coverage in Compustat. A firm is classified as mature if it is more than 15 years after the year it first appears in Compustat with a stock price. The high-tech industries are SICs 283, 357, 366, 367, 382, 384, and 737. The heavy line plots the sum of R&D for all mature high-tech firms, the dashed line plots the sum of gross cash flow, and the thin line plots the sum of net new stock issues with negative net issues set equal to zero.

Read:

- For all high-tech firms, R&D, cash flow, and new stock issues all surge in the late 1990s.
- For *young* firms, the co-movement is especially stark: R&D rises along with cash flow and equity issuance, then falls when both collapse after 2000.
- For *mature* firms, R&D grows much more smoothly, and stock issuance is tiny.

Economic message. These figures are the descriptive heart of the paper. They show why the authors focus on young firms and why external equity is essential to the story.

5.3 Figures 3 and 4: the cycle is a young-firm cycle

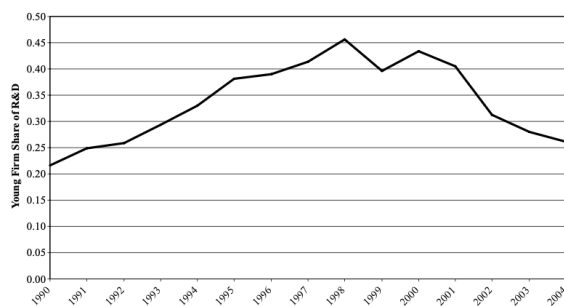


Figure 3. Young firm share of total R&D (regression sample). The line plots the share of regression sample R&D accounted for by young firms over time. A firm is classified as young for the first 15 years following the year it first appears in Compustat with a stock price. The regression sample is described in Section III.A of the paper.

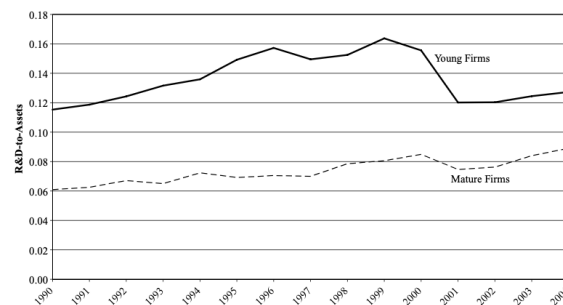


Figure 4. Median R&D-to-assets ratios (regression sample). The solid line plots the median R&D-to-assets ratio for young firms in the regression sample over time, and the dashed line plots the median R&D-to-assets ratio for mature firms. A firm is classified as young for the first 15 years following the year it first appears in Compustat with a stock price, and mature thereafter. The regression sample is described in Section III.A of the paper.

Read:

- Figure 3 shows that the young-firm share of sample R&D rises strongly through the late 1990s and peaks at 45.7% in 1998.
- Figure 4 shows that the median R&D-to-assets ratio of young firms rises substantially and then drops after 2000.
- Mature-firm R&D intensity is far smoother and follows a modest upward trend.

Economic message. The aggregate boom is driven both by composition (young firms become a larger share of total R&D) and by within-young-firm behavior (their own R&D intensity becomes cyclical).

5.4 Table I: why young firms should be financially constrained

Read:

- Young firms have much higher R&D intensity than mature firms.
- Young firms' mean stock-issue ratio (0.268) is larger than their mean cash-flow ratio (0.217), whereas mature firms issue almost no new stock.
- Debt is quantitatively unimportant for both groups.
- The financing shares show that public equity is a major marginal source of funds for young firms and essentially irrelevant for mature firms.

Economic message. The regression results are already foreshadowed in the summary statistics: young firms have to raise equity, mature firms usually do not.

Table I
Sample Descriptive Statistics

The regression sample is constructed from publicly traded high-tech firms with coverage in the Compustat database during 1990 to 2004. We exclude firms incorporated outside of the United States, firms with no stock price data, and firms without at least six R&D observations. We also exclude firms if the sum of their cash flow-to-assets ratio over the sample period is less than or equal to zero. All variables are scaled by beginning-of-period total assets. Outliers in all variables are trimmed at the 1% level. Young-firm observations are those less than or equal to 15 years from the initial appearance of a stock price in Compustat; mature-firm observations are more than 15 years from the appearance of a stock price. The final column reports p -values for tests that the mean and median values differ across young and mature firms. By three-digit SIC code, the high-tech industries are: 283, 357, 366, 367, 382, 384, and 737.

Variable and Statistic	Full Sample	Young Firms	Mature Firms	Difference (p -value)
rd_t				
Mean	0.170	0.194	0.098	0.000
Median	0.116	0.137	0.074	0.000
SD	0.217	0.240	0.100	
90 th percentile	0.350	0.395	0.186	
$capex_t$				
Mean	0.064	0.069	0.049	0.000
Median	0.041	0.042	0.039	0.000
SD	0.149	0.171	0.042	
90 th percentile	0.130	0.143	0.102	
s_t				
Mean	1.212	1.227	1.167	0.000
Median	1.083	1.076	1.099	0.145
SD	0.993	1.095	0.597	
90 th percentile	2.122	2.213	1.837	
cf_t				
Mean	0.205	0.217	0.172	0.000
Median	0.185	0.194	0.166	0.000
SD	0.369	0.398	0.261	
90 th percentile	0.457	0.495	0.347	
stk_t				
Mean	0.204	0.268	0.021	0.000
Median	0.006	0.010	0.001	0.000
SD	1.160	1.338	0.155	
90 th percentile	0.427	0.643	0.050	

(continued)

Table I—Continued

Variable and Statistic	Full Sample	Young Firms	Mature Firms	Difference (p -value)
dbt_t				
Mean	0.009	0.009	0.007	0.515
Median	0.000	0.000	0.000	0.655
SD	0.344	0.394	0.113	
90 th percentile	0.061	0.058	0.067	
Sum cash flow/net finance				
Mean	0.686	0.659	0.916	0.000
Median	0.731	0.692	0.957	0.000
SD	0.399	0.421	0.627	
90 th percentile	1.139	1.110	1.357	
Sum new stock/net finance				
Mean	0.289	0.320	0.068	0.000
Median	0.219	0.247	0.027	0.000
SD	0.367	0.381	0.419	
90 th percentile	0.807	0.850	0.484	
Sum new debt/net finance				
Mean	0.021	0.015	0.019	0.820
Median	0.000	−0.001	0.000	0.002
SD	0.182	0.194	0.382	
90 th percentile	0.229	0.230	0.378	

5.5 Tables II and III: pooled significance is driven by young firms

Read:

- Table II shows that in the pooled sample, contemporaneous cash flow and stock issuance both enter positively and significantly.
- Table III reveals where that pooled effect comes from: young firms.
- For mature firms, cash flow and stock issuance coefficients are economically small and statistically insignificant.
- For young firms, current cash flow is about 0.15–0.17 and current stock issuance is about 0.148, with little change when industry-year dummies are added.

Economic message. The age split is not a robustness footnote. It is the paper's core identification device.

5.6 Table IV: the core result is robust

Read:

- Adding $t - 2$ instruments does not materially change the results.
- Orthogonal-deviations GMM also leaves the finance coefficients broadly unchanged.
- Young-firm cash-flow and stock-issue coefficients remain large and significant; mature-firm coefficients remain small.

Table II Dynamic R&D Regressions: Pooled Results				
Dynamic R&D regressions on the sample of high-tech firms described in Table I are estimated with one-step GMM in first differences to eliminate firm effects. The dependent variable is $rd_{i,t}$. Variable definitions with Compustat data codes are provided in the Appendix. Level variables dated $t-3$ and $t-4$ are used as instruments. Aggregate or industry-specific year dummies are included in all regressions (as indicated in the table). Standard errors robust to heteroskedasticity and within-firm serial correlation appear below point estimates. The statistics $m1$ and $m2$ test the null of no first- and second-order autocorrelation in the first-differenced residuals. The χ^2 statistic is a Chi-square test of the null that the sum of the current and lagged cash flow coefficients is zero; atk χ^2 is the corresponding test for stock issues. Sargan is a test of the null that that the overidentifying restrictions are valid.				
Variable	Baseline Euler Equation		Euler Equation with Financial Variables	
$rd_{i,t-1}$	0.808 (0.215)	0.789 (0.200)	0.380 (0.134)	0.403 (0.130)
$rd_{i,t-1}^2$	-0.409 (0.139)	-0.384 (0.130)	-0.145 (0.076)	-0.153 (0.073)
s_t			-0.020 (0.018)	-0.007 (0.015)
s_{t-1}	-0.007 (0.016)	-0.013 (0.017)	-0.015 (0.009)	-0.022 (0.009)
cf_t			0.170 (0.041)	0.158 (0.040)
cf_{t-1}	-0.015 (0.045)	-0.004 (0.045)	0.061 (0.016)	0.066 (0.016)
atk_t			0.151 (0.017)	0.149 (0.017)
atk_{t-1}			-0.018 (0.004)	-0.017 (0.004)
Time Dummies	Aggregate	Industry Level	Aggregate	Industry Level
$m1$ (p-value)	0.000	0.000	0.000	0.000
$m2$ (p-value)	0.401	0.311	0.249	0.345
χ^2 χ^2 (p-value)			0.000	0.000
atk χ^2 (p-value)			0.000	0.000
Sargan (p-value)	0.002	0.889	0.138	1.000
Observations	12,248	12,248	12,224	12,224

Table III Dynamic R&D Regressions for Separate Young- and Mature-Firm Samples				
Dynamic R&D regressions on the samples of young and mature high-tech firms described in Table I are estimated with one-step GMM in first differences to eliminate firm effects. The dependent variable is $rd_{i,t}$. Variable definitions with Compustat data codes are provided in the Appendix. Level variables dated $t-3$ and $t-4$ are used as instruments. Aggregate or industry-specific year dummies are included in all regressions (as indicated in the table). Standard errors robust to heteroskedasticity and within-firm serial correlation appear below point estimates. The test statistics reported at the bottom of the table are described in the caption for Table II.				
Variable	Mature Firms		Young Firms	
$rd_{i,t-1}$	0.632 (0.170)	0.628 (0.172)	0.382 (0.137)	0.392 (0.130)
$rd_{i,t-1}^2$	-0.831 (0.277)	-0.830 (0.226)	-0.142 (0.077)	-0.144 (0.074)
s_t	0.049 (0.016)	0.046 (0.015)	-0.023 (0.020)	-0.006 (0.019)
s_{t-1}	-0.031 (0.011)	-0.031 (0.010)	-0.015 (0.010)	-0.023 (0.009)
cf_t	-0.015 (0.032)	-0.005 (0.031)	0.166 (0.044)	0.150 (0.043)
cf_{t-1}	0.041 (0.020)	0.040 (0.018)	0.003 (0.017)	0.012 (0.017)
atk_t	0.031 (0.029)	0.034 (0.030)	0.148 (0.017)	0.148 (0.017)
atk_{t-1}	-0.010 (0.011)	-0.010 (0.011)	-0.012 (0.004)	-0.017 (0.004)
Time Dummies	Aggregate	Industry Level	Aggregate	Industry Level
$m1$ (p-value)	0.000	0.000	0.000	0.000
$m2$ (p-value)	0.795	0.880	0.294	0.432
χ^2 χ^2 (p-value)	0.382	0.212	0.000	0.000
atk χ^2 (p-value)	0.339	0.470	0.000	0.000
Sargan (p-value)	0.633	1.000	0.103	1.000
Observations	3,393	3,393	8,831	8,831

Table IV Dynamic R&D Regressions for Separate Young- and Mature-Firm Samples: Robustness				
Dynamic R&D regressions corresponding to the first-differenced GMM results in Table III are estimated with alternative data transformations (first differences and orthogonal deviations) and instrument sets. The dependent variable is $rd_{i,t}$. Variable definitions with Compustat data codes are provided in the Appendix. One-step GMM estimates using first differences to remove firm effects are presented with level variables dated $t-2$ to $t-4$ used as instruments. GMM estimates using a forward orthogonal deviations transformation to remove firm effects are presented with level variables dated $t-1$ to $t-3$ used as instruments. Industry-specific year dummies are included in all regressions. Standard errors robust to heteroskedasticity and within-firm serial correlation appear below point estimates. The test statistics reported at the bottom of the table are described in the caption for Table II.				
Variable	Mature Firms		Young Firms	
	First Differences with $t-2$ to $t-4$ Instruments	Orthogonal Deviations with $t-1$ to $t-3$ Instruments	First Differences with $t-2$ to $t-4$ Instruments	Orthogonal Deviations with $t-1$ to $t-3$ Instruments
$rd_{i,t-1}$	0.600 (0.152)	0.556 (0.104)	0.403 (0.116)	0.411 (0.122)
$rd_{i,t-1}^2$	-0.785 (0.263)	0.077 (0.130)	-0.151 (0.060)	-0.147 (0.069)
s_t	0.035 (0.014)	0.046 (0.009)	-0.004 (0.016)	0.015 (0.014)
s_{t-1}	-0.028 (0.009)	-0.035 (0.007)	-0.023 (0.008)	-0.029 (0.008)
cf_t	0.019 (0.027)	0.004 (0.021)	0.156 (0.034)	0.175 (0.037)
cf_{t-1}	0.039 (0.017)	0.015 (0.015)	0.008 (0.016)	-0.016 (0.016)
atk_t	0.043 (0.021)	0.051 (0.032)	0.138 (0.015)	0.131 (0.014)
atk_{t-1}	-0.011 (0.012)	-0.040 (0.016)	-0.017 (0.004)	-0.016 (0.005)
Time Dummies	Aggregate	Industry Level	Aggregate	Industry Level
$m1$ (p-value)	0.000	0.027	0.000	0.000
$m2$ (p-value)	0.733	0.207	0.421	0.306
χ^2 χ^2 (p-value)	0.035	0.374	0.000	0.000
atk χ^2 (p-value)	0.228	0.000	0.000	0.000
Sargan (p-value)	1.000	0.427	0.975	0.983
Observations	3,393	3,393	8,831	8,822

Economic message. The finance–R&D link is not a fragile feature of one estimator or one instrument set.

6 Short summary

- The paper studies whether finance-supply shifts can explain the 1990s U.S. R&D boom.
- It focuses on seven high-tech industries where young public firms rely heavily on equity finance and debt is nearly absent.
- Using a dynamic panel R&D model estimated by GMM, it shows that cash flow and new stock issues strongly predict R&D for young firms but not for mature firms.
- The equality in magnitude of the cash-flow and stock-issue coefficients supports the interpretation that both are funding sources for constrained firms rather than unrelated demand proxies.
- The estimated young-firm coefficients are large enough to explain most of the R&D boom from 1994 to 2000 and most of the decline after 2000.
- The results survive many alternatives: Tobin’s q , industry-year effects, alternative age cutoffs, bond-rating and payout splits, alternative GMM implementations, and checks against overinvestment or investor-catering interpretations.
- The overall implication is that equity-market conditions are a first-order determinant of innovative activity for young high-tech firms.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that financing constraints are economically important for R&D among young high-tech public firms, and that shifts in the supply of internal and external equity finance can explain a large part of the late-1990s U.S. R&D boom and the subsequent bust.

7.2 Economic intuition

Young high-tech firms have little collateral, little debt capacity, and large R&D needs. When cash flow rises or equity becomes easier to issue, these firms move closer to their desired R&D path. Mature firms, by contrast, usually already have enough internal funds, so the same finance shocks do not change their R&D very much.

7.3 Takeaway

For innovation-intensive sectors, finance matters not only through banks and credit markets but also through stock markets and internal equity. If one wants to understand the link between finance and growth, the funding of young innovative firms has to be at the center of the story.